

Cyber Range

Toolset Reference Guide

Scenario-Mapped Tools for Red Team · SOC · Blue Team Training
Three Core Scenarios · Eight Additional Scenarios · 20 Reusable Platform Modules

Scenario W1 — WannaCry-Style SMB Worm

Tools mapped per attack stage across Red Team, SOC, and Blue Team functions. Referenced throughout the WannaCry scenario.

Red Team Tools

Stage	Tool
Host Discovery	Nmap
SMB Enumeration	CrackMapExec
Exploit Simulation	EternalBlue Simulator
Payload Deployment	Payload Dropper Simulator
Persistence	Windows Service Creator
Kill Switch Check	DNS Query Simulator
Lateral Movement	SMB Propagation Engine
Recovery Deletion	Shadow Copy Deletion Simulator
Encryption	Ransomware Engine

SOC Tools

Stage	Tool
Scan Detection	Zeek
IDS Detection	Suricata
Traffic Analysis	Wireshark
Log Analysis	Splunk
Endpoint Monitoring	Sysmon
IOC Search	Threat Hunting Console

Blue Team Tools

Stage	Tool
Host Isolation	EDR Quarantine

SMBv1 Detection	Vulnerability Scanner
Network Segmentation	Firewall Manager
Kill Switch Sinkhole	DNS Manager
Patch Deployment	WSUS
Backup Restore	Veeam Simulator

Scenario R5 — Phishing-to-Encrypt Ransomware

Tools mapped directly from the R5 scenario design, covering the full kill chain from initial phishing to encryption and recovery.

Red Team Tools

Stage	Tool
Email Creation	GoPhish
Attachment Creation	Office Macro Simulator
PowerShell Execution	PowerShell Engine
Persistence	Scheduled Task Creator
Process Injection	Beacon Simulator
C2	HTTPS Beacon Simulator
Discovery	BloodHound
Lateral Movement	RDP Simulator
Encryption	Ransomware Simulator

SOC Tools

Stage	Tool
Email Analysis	Secure Email Gateway
Process Tree	Sysmon
EDR Alerts	CrowdStrike/Defender Simulator
PowerShell Monitoring	Event ID 4104 Viewer
Timeline Analysis	SIEM Timeline
IOC Investigation	Threat Hunting Module

Blue Team Tools

Stage	Tool
Host Isolation	Quarantine Console
Disable User	Active Directory

Password Reset	IAM Console
Block C2	Firewall/DNS Block
Restore Backup	Backup Console
Incident Declaration	IR Management Console

Scenario C5 — EDR Outage Exploitation

Core tools explicitly needed when EDR visibility is lost. Covers opportunistic exploitation of monitoring blind spots through to full ransomware deployment.

Red Team Tools

Stage	Tool
OSINT Monitoring	Shodan
Password Spray	Hydra Simulator
VPN Access	VPN Gateway
Persistence	Scheduled Task / Service Creator
Credential Theft	Credential Dump Simulator
AD Enumeration	BloodHound
Lateral Movement	PSEXec / RDP
Data Collection	File Collector
Exfiltration	Rclone Simulator
Shadow Copy Deletion	Recovery Disable Simulator
Ransomware	Enterprise Encryptor Simulator

SOC Tools

Stage	Tool
Authentication Monitoring	Azure AD Logs
VPN Monitoring	VPN Log Analyzer
NetFlow Analysis	Zeek
Proxy Monitoring	Proxy Analyzer
DLP Monitoring	DLP Console
File Audit Review	File Server Audit Dashboard
Incident Timeline	SIEM

Blue Team Tools

Stage	Tool
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MFA Enforcement	Azure MFA
Account Disable	Active Directory
Network Segmentation	NAC Controller
Alternate Monitoring	Sysmon
Memory Analysis	Volatility
Backup Verification	Backup Dashboard
Offline Backup Isolation	Backup Vault
Full Host Isolation	Incident Response Console

Additional Scenarios — Recommended Builds

The most frequently requested cyber range scenarios after the three core scenarios above. Each maps to the same Red Team / SOC / Blue Team structure.

Scenario	Red Team Tools	SOC Tools	Blue Team Tools
Insider Threat	USB Simulator, File Copier	UEBA	DLP
Business Email Compromise	GoPhish	Email Gateway	MFA
Cloud Breach	AWS CLI Simulator	CloudTrail	IAM
Active Directory Attack	BloodHound	SIEM	AD Hardening
Data Exfiltration	Rclone	DLP	Firewall
Supply Chain Attack	Software Repository Simulator	SIEM	Software Integrity Monitoring
Web Server Compromise	Burp Suite Simulator	WAF Logs	WAF
Ransomware Double Extortion	Exfiltration Simulator	DLP	Backup Recovery

Core Platform Tool Library — 20 Reusable Modules

Rather than creating new tools for every scenario, build these reusable modules once. With these 20 core modules you can support nearly every MITRE ATT&CK-based Red Team, Blue Team, and SOC training scenario.

Module	Module
1. Email Simulator	2. PowerShell Simulator
3. Active Directory Simulator	4. RDP Simulator
5. SMB Simulator	6. VPN Simulator
7. DNS Simulator	8. HTTP/HTTPS Traffic Simulator

9. C2 Simulator	10. Ransomware Simulator
11. Exfiltration Simulator	12. SIEM Console
13. EDR Console	14. Threat Hunting Console
15. Incident Response Console	16. Backup Recovery Console
17. Firewall Console	18. IAM/MFA Console
19. Asset Management Console	20. Scoring Engine

End of Cyber Range Toolset Reference